Traffic Speed Forecasting

Independent-Sensor Prediction with Traditional Machine Learning

FACULTY-ALLOCATED PROJECT

Multi-Agent Traffic Flow Control via Graph Attention Networks and LWR Models

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Scope of this report

Phase 1 evaluates speed forecasting using traditional ML. Graph Attention Networks, calibrated LWR modelling and traffic control remain future work. Any future control experiment will be simulated only.

Abstract

This study evaluates traditional machine-learning methods for short-term traffic speed forecasting on METR-LA. The inspected dataset contains 34,272 five-minute timestamps and 207 sensors. Persistence and a training-only historical average are compared with independent-sensor Random Forest and XGBoost models using 38 causal features.

Chronological partitions, purged target boundaries, training-only fitted statistics and common target masks support a controlled comparison. On the previously inspected test interval, XGBoost achieves pooled mean absolute errors of 2.920, 3.376 and 3.874 mph at 15, 30 and 60 minutes, representing reductions of 16.1%, 19.1% and 27.2% versus persistence. Random Forest has lower pooled root mean squared error at those horizons and lower 60-minute peak-period MAE.

The results are exploratory because the test interval was exposed during earlier prototype work. The study establishes reproducible temporal baselines for future graph-aware experiments, without claiming a demonstrated graph-model or traffic-control benefit.

SENSORS	INPUT FEATURES	60-MINUTE MAE
207	**38**	**3.874**
METR-LA network	Causal temporal features	mph / XGBoost

Keywords: traffic forecasting, METR-LA, regression, Random Forest, XGBoost, chronological validation, missing data.

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1 Introduction

Traffic conditions vary across time and locations. Short-term speed forecasts can support congestion analysis and future simulated traffic-management studies. A forecasting experiment first needs to establish whether learned models improve on simple references under comparable conditions.

The broader project concerns network-aware prediction and simulated control. Phase 1 predicts each sensor's future speed from its own recent readings and known calendar time. The verified road graph provides dataset context and a foundation for later experiments; it is not an input to the current estimators.

2 Problem Statement and Objectives

For sensor i, forecast origin t and horizon h, predict $v_{t+h,i}$, for $h = 1, \ldots, 12$. One step is five minutes; headline horizons $h = 3, 6, 12$ correspond to 15, 30 and 60 minutes.

- Validate the speed matrix, chronology and sensor alignment.
- Examine training-period patterns and missing observations.
- Construct causal features with training-only fitted preprocessing.
- Compare two reference methods and two learned model families.
- Evaluate common-mask errors across horizons and traffic periods.
- Preserve reproducible evidence and identify limitations.

Prediction target

This is speed regression. Speed is not interchangeable with measured flow, density or occupancy. Classification accuracy is not an applicable score.

3 Literature Review

3.1 Tree ensembles

Random Forests combine randomized trees and average predictions for regression [1]. They motivate a nonlinear baseline using engineered temporal features. Traffic-specific performance must be measured rather than inferred from the general algorithm.

XGBoost is a scalable tree-boosting system [2]. Here, direct regressors use temporal features and a small predefined configuration search. Boosting is not assumed to dominate every error metric.

3.2 Network-aware forecasting

DCRNN models directed-network and temporal dependencies [3]. Its repository supplies benchmark data instructions and graph context [4]. Our purged partitions differ from published protocols, so published scores are not directly ranked against this experiment.

Graph Attention Networks learn weighted aggregation over graph neighbourhoods [5]. They motivate Phase 2 but are not trained here. The literature supports testing spatial information, rather than assuming it will improve the current baseline.

4 Dataset and Data Validation

4.1 Dataset description

The inspected METR-LA HDF5 matrix contains one measured channel: speed. Units follow the benchmark source's mph description; numerical values alone do not independently establish physical units. Complete-file diagnostics describe the supplied data; they do not imply full-file fitting or training EDA.

Table 1: Properties measured from the inspected dataset.

Property	Observed Value
Timestamps / sensors	34,272 / 207
Recorded interval	1 March–27 June 2012
Sampling interval	Five minutes
Full-file missing fraction	8.11%
Training-fit missing fraction	7.13%
Valid observed speeds	Approximately 0.333–70 mph
Timezone	Not encoded in the inspected timestamp index

4.2 Integrity and missing-value policy

Checks cover unique five-minute timestamps, valid raw values, sensor ordering and graph alignment. Negative or infinite raw speeds are rejected. Adjacency rebuilt from supplied distances matches the reference DCRNN adjacency.

Zeros and NaNs are treated as missing under the benchmark policy. This is not proof that every physical zero-speed reading means a detector outage. Raw-data redistribution rights remain unverified separately from the upstream code licence.

5 Exploratory Data Analysis

Training-period EDA examines speed distributions, sensor summaries, missingness, daily profiles and weekday/weekend differences.

- Weekday speed reductions around morning and evening commuting periods motivate cyclic time features and peak-period reporting.
- Missing blocks and long sensor-specific gaps motivate masks and missingness flags. The audit records 1,301 all-sensor-missing steps in the training fit interval. Their physical cause is not established.
- Differences among sensors motivate per-sensor error analysis.
- Within-day variation motivates lags, rolling summaries and differences.

These findings motivate design choices; they do not establish a measured feature-ablation benefit.



Figure 1: Training-period mean speeds by recorded clock hour for weekdays and weekends. Timezone interpretation remains unresolved.



Figure 2: Training-period missingness by sensor and day under the stated zero/NaN convention.

6 Preprocessing and Feature Engineering

6.1 Causal history and imputation

Each sample uses observations $t-11,\ldots,t$: 12 five-minute sampling slots whose first and last timestamps are 55 minutes apart. Only information available at origin t is used.

Missing inputs are forward-filled from earlier observations within the sample's history. Leading gaps use the sensor's training median. Backward fill is not used. Missing targets are never imputed and are excluded through the shared evaluation mask. Cyclic features for time index u with period P are

$$\sin(2\pi u/P), \qquad \cos(2\pi u/P).$$

They represent proximity across cycle boundaries.

Table 2: The 38 input features per sensor and forecast origin.

Feature Group	Count
Speed lags, latest reading through lag 11	12
Original missingness flag for each lag	12
Means over the last 3, 6 and 12 readings	3
History standard deviation, minimum and maximum	3
One-step and history-endpoint differences	2
History missing fraction	1
Sine/cosine of time of day and time of week	4
Weekend indicator	1
Total	**38**

6.2 Scaling and feature selection

No scaler is used: this threshold-based tree implementation does not require standardization. Medians and historical-average tables are fitted on training observations only.

All 38 features are retained. No formal feature selection, PCA or feature-removal ablation is claimed. Feature importance supports descriptive interpretation; correlated inputs and model-specific importance definitions limit causal interpretation and cross-model comparison.

7 Experimental Design

7.1 Chronology and boundary purging

Nominal origin proportions are 70% training, 10% validation and 20% test. Validation is divided into tuning and second-validation blocks. The latter is called `val_stack` for historical compatibility; no stacking model is implemented.

The last 11 origins of each earlier block are removed so its targets do not extend beyond the next block's first origin. Previously observed input history may cross boundaries.

Table 3: Retained forecast origins after purging.

Partition	Origins
Training	23,963
Tuning validation	1,701
Second validation	1,702
Test	6,850

7.2 Training and hyperparameter selection

Tree training uses every second retained training origin to bound computation. Evaluation uses all retained evaluation origins. Two predefined candidates per learned family are compared on six fixed pilot sensors using pooled valid-point MAE at 15, 30 and 60 minutes on tuning validation only. Both XGB candidates use minimum child weight 10, row

Table 4: Pilot-sensor tuning results.

Candidate	MAE
RF: 48 trees, depth 12, minimum leaf 8, feature fraction 0.8; selected	3.829
RF: 64 trees, depth 16, minimum leaf 5, feature fraction 0.8	3.881
XGB: 100 trees, depth 3, learning rate 0.08, squared error	3.824
XGB: 140 trees, depth 3, learning rate 0.06, absolute error; selected	3.413

subsampling 0.8 and column subsampling 0.9. The random seed is 42. Selected settings are then fitted across sensors. Pilot scores are not full-sensor validation results or proof of global optimality.

Evaluation status

Earlier prototype work inspected the test interval. Current selection uses tuning validation, but the test is not an untouched holdout. Results are exploratory; stronger generalization claims require a prospectively agreed additional evaluation protocol.

8 Models and Theoretical Foundations

Four methods are compared: two references and two learned model families. Historical identifiers are M0, M1, M3 and M4.

8.1 M0: Persistence

Repeat the latest valid input reading for every horizon:

$$\widehat{v}_{t+h,i} = v_{t,i}^{\text{last valid}}.$$

An entirely missing history uses the historical-average fallback. Persistence is a useful reference when conditions change slowly.

8.2 M1: Historical average

Average valid training readings by sensor and time-of-week slot. Writing $s(t+h)$ for the target slot,

$$\widehat{v}_{t+h,i} = \text{mean}\{v_{\tau,i} : \tau \in \mathcal{T}_{\text{train}},\ s(\tau) = s(t+h),\ v_{\tau,i} \text{ valid}\}.$$

Unobserved slots fall back to a training time-of-day average and then the sensor training mean. Predictions may vary with horizon because the target slot changes, even if pooled errors are nearly flat.

8.3 M3: Multi-output Random Forest

One forest per sensor jointly predicts all 12 future speeds. For B vector-output trees,

$$\widehat{\boldsymbol{v}}_{t,i} = \frac{1}{B}\sum_{b=1}^{B} T_{b,i}(\boldsymbol{x}_{t,i}).$$

Bootstrap and feature subsampling introduce diversity. Regression predictions are averaged, not voted. Training rows with any missing future label are excluded because the complete target vector is required. Bootstrap sampling inside training does not replace chronological evaluation.

8.4 M4: Direct XGBoost

A separate regressor is fitted per sensor and horizon:

$$\widehat{v}_{t+h,i} = \sum_{k=1}^{K} \eta f_{k,i,h}(\boldsymbol{x}_{t,i}).$$

A schematic regularized objective is

$$\mathcal{L} = \sum_{j\in\mathcal{J}_{i,h}} \ell(y_j, \widehat{y}_j) + \sum_{k} \Omega(f_k),$$

where $\mathcal{J}_{i,h}$ contains valid labels for that horizon. The selected candidate uses absolute-error loss. Only the relevant horizon's invalid labels are excluded. Insufficient labels trigger the predefined historical-average fallback; fallback activity is recorded.

9 Evaluation Methodology

For n common valid sensor–time targets, let $e_j = \widehat{v}_j - v_j$:

$$\mathrm{MAE} = \frac{1}{n}\sum_{j=1}^{n} |e_j|, \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n} e_j^2}, \tag{2}$$

$$\mathrm{MAPE} = \frac{100}{n}\sum_{j=1}^{n} |e_j / v_j|. \tag{3}$$

MAE and RMSE are in mph; MAPE is percentage error. All methods share valid-target masks. Zero/missing targets are excluded, but low positive speeds can still make MAPE unstable. Neither these errors nor 100 − MAPE is classification accuracy.

Metrics pool valid points rather than equally averaging sensors. Peak targets are weekdays 07:00–10:00 and 16:00–19:00 in recorded clock time. Other targets are off-peak; timezone interpretation remains provisional.

10 Results and Model Comparison

Scores come from saved evidence in `verification/full-phase1-20261026/`. Writing the report did not retrain or rescore models.

10.1 Headline results

Table 5: Exploratory test MAE, mph; lower is better.

Method	15 min	30 min	60 min
Persistence	3.482	4.173	5.321
Historical average	4.187	4.187	4.187
Random Forest	2.972	3.423	3.896
XGBoost	**2.920**	**3.376**	**3.874**

Table 6: Exploratory test RMSE, mph; lower is better.

Method	15 min	30 min	60 min
Persistence	6.374	7.983	10.250
Historical average	7.852	7.852	7.852
Random Forest	**5.438**	**6.374**	**7.255**
XGBoost	5.777	6.911	7.993

Table 7: Exploratory test MAPE, percent; lower is better.

Method	15 min	30 min	60 min
Persistence	8.57	10.77	14.39
Historical average	13.03	13.03	13.03
Random Forest	8.07	**9.81**	**11.71**
XGBoost	**8.04**	10.06	12.59

XGB has the lowest headline pooled MAE, improving on persistence by 16.1%, 19.1% and 27.2%. Both learned families improve headline pooled MAE over both references. This

does not hold for every metric: 60-minute XGB RMSE exceeds historical average's RMSE. RF has the lowest headline pooled RMSE.

Rankings vary by interval: second-validation 60-minute MAE is 3.746 mph for RF versus 3.765 mph for XGB.

Saved figure not embedded. Upload `mae_horizons_test.png` into `images/`.

Figure 3: Exploratory test-period MAE across all 12 forecast horizons.

10.2 Traffic-period comparisons

Table 8: Exploratory 60-minute MAE, mph, by target-period slice.

Method	Peak	Off-peak	Weekday	Weekend
Persistence	7.776	4.704	5.711	3.801
Historical average	6.528	3.598	4.363	3.499
Random Forest	**5.954**	3.379	**4.041**	3.333
XGBoost	6.308	**3.261**	4.073	**3.098**

XGB peak MAE is approximately 1.93 times its off-peak MAE. RF has lower 60-minute peak and weekday MAE; XGB has lower off-peak and weekend MAE. One pooled score does not capture every trade-off.

11 Error Analysis and Discussion

At 60 minutes, XGB improves per-sensor MAE over persistence for 206 of 207 sensors and over RF for 120 of 207 sensors. The unweighted mean sensor-level reduction versus persistence is approximately 1.449 mph. This weighting differs from pooled valid-point MAE. One sensor worsens by approximately 0.104 mph; signed improvement maps must display this regression.

A fixed sensor/day example illustrates local behaviour, not network-wide performance. Feature importances describe fitted model usage rather than causal effects or feature-removal benefits. The estimators omit neighbouring sensors' observations. Spatial error

Saved figure not embedded. Upload `forecast_example.png` into `images/`.

Figure 4: Saved actual speeds and 60-minute forecasts for one fixed sensor and test day. The example is illustrative.

patterns motivate a Phase 2 hypothesis, not proof that missed congestion propagation caused errors. Matched evaluation and a no-neighbour ablation are needed to distinguish graph information from capacity or training differences.

12 Implementation and Reproducibility

```
DLML_Midsem_Project/
  configs/       data and model settings
  src/trafficml/ loaders, features, models and metrics
  scripts/       training, evaluation and inference
  tests/         correctness and integrity checks
  verification/  saved evidence and manifests
  report/        generated results and report sources
  slides/        presentation artifacts
  docs/          methods and contribution records
```

The recorded run used Python 3.12.14, pandas 2.2.3, scikit-learn 1.7.2 and XGBoost 3.1.1. Full pins are in `requirements.txt`. Source, data and configuration hashes identify experiment inputs. Original run metadata retains its historical workspace path.

With the documented environment and data available:

```
python scripts/check_data.py
pytest -q -ra
ruff check .
ruff format --check .
python scripts/run_all.py \
  --run-dir results/runs/new-experiment
```

These are reproduction instructions, not newly executed experiments for this report. The last command trains models and creates artifacts; it is not just a report viewer.

Preserve `report/phase1/results.md`, `report/phase1/metrics.tex` and `report/phase1/run_reference.json` alongside the narrative. Regenerate evidence and synchronize report tables when results change.

13 Verification

The integration record documents 56 real-data tests passing, lint/format checks, package building, source-fingerprint validation and saved-model inference. These are recorded results, not checks rerun by compiling this report. Tests cover data/graph integrity, causal features, purged partitions, common metric masks and artifact provenance. Passing checks does not certify faculty acceptance or individual understanding.

14 Limitations

- The previously exposed test interval makes results exploratory.
- Data contain speed only, for one region over four months.
- Two candidates per family and six pilot sensors are a limited tuning budget.
- Stride-two training and RF/XGB label-exclusion differences affect comparisons.
- Timezone/DST and raw-data redistribution rights remain unresolved.
- The raw XGB test tensor contains 24 negative values among 17,015,400 forecasts across all horizons. They were retained rather than clipped after observing scores.
- MAPE is sensitive near zero. Overlapping horizons and repeated observations are dependent; large counts do not establish significance.
- Project-specific faculty acceptance requires separate confirmation.

15 Phase 2 Plan

Evaluate a graph-aware model using the verified adjacency and a documented common protocol. Include temporal-only and no-neighbour variants. Prospectively agree how to handle the already-exposed evaluation interval.

Investigate horizon, traffic-period and sensor-level trade-offs rather than assuming improvement. A calibrated LWR analysis needs justified flow/density information and assumptions beyond this speed matrix. Any future signal-policy experiment must remain simulated. No graph, LWR or control result is claimed in Phase 1.

16 Conclusion

Phase 1 establishes an independent-sensor speed-forecasting pipeline with data checks, causal features, chronological validation and common masked evaluation. XGB has the lowest exploratory headline pooled MAE; RF has lower headline pooled RMSE and lower 60-minute peak MAE. These measured temporal baselines and limitations provide a starting point for graph-aware experiments.

17 Team Contributions

Member	Recorded Scope
Kavya Katal	Project coordination and preparation; remaining environment/configuration, orchestration, evaluation, inference, CI, evidence and academic artifacts; combined verification and report preparation.
Rudraksh Sharma	Data provenance, loading and graph-integrity handoff; integration and review record.
Vridhi Chaudhary	Missing-data handling, splits and temporal-feature handoff; integration and review record.
Anshuman Mehta	Dataset EDA and interpretation handoff; integration and review record.
Bineet Keshari	Baselines, shared tree estimators and error-analysis handoff; integration and review record.

Original prototype provenance, member handoff records and merged pull-request history remain part of the project record. Individual understanding and reproduction must be demonstrated separately by each member.

References

[1] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, pp. 5–32, 2001. https://doi.org/10.1023/A:1010933404324.

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[3] Y. Li, R. Yu, C. Shahabi and Y. Liu, "Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting," *ICLR*, 2018. https://arxiv.org/abs/1707.01926.

[4] Y. Li et al., "DCRNN: Implementation and data instructions," project repository. https://github.com/liyaguang/DCRNN.

[5] P. Veličković et al., "Graph Attention Networks," *ICLR*, 2018. https://arxiv.org/abs/1710.10903.

A Evidence Register

Component	Repository Evidence
Dataset and EDA	verification/full-phase1-20261026/analysis/dataset_audit.json; training plots and sensor audit in the same directory.
Features and theory	docs/METHODS_AND_VIVA.md; src/trafficml/features/temporal.py; src/trafficml/models/.
Splits and settings	verification/full-phase1-20261026/manifest.json; verification/full-phase1-20261026/tuning.json.
Measured results	verification/full-phase1-20261026/analysis/all_metrics.csv; verification/full-phase1-20261026/leaderboard_test.csv.
Sensor analysis	verification/full-phase1-20261026/analysis/per_sensor_mae_60.csv; verification/full-phase1-20261026/analysis/feature_importance.csv.
Verification	verification/integration-20261027/; tests/.
Contributions	docs/CONTRIBUTIONS.md; member review logs and merged PRs.

B Phase 1 Requirement Mapping

Requirement	Coverage
Literature	Algorithm references and graph-model background.
Dataset and EDA	Integrity checks, missingness and temporal patterns.
Features	Causal imputation and 38 documented inputs.
Theory	Forecast formulation, models and metric equations.
Model application	References, learned models and validation tuning.
Code quality	Reproduction workflow and recorded verification.
Report	This narrative and the linked evidence.
Presentation and viva	Separate preparation, rehearsal and individual demonstration required.

C Figure Files and Formatting

The `images/` folder contains saved `daily_profile.png`, `missingness.png`, `mae_horizons_test.png` and `forecast_example.png` from `verification/full-phase1-20261026/analysis/`. Missing images produce labelled placeholders during drafting; replace every placeholder before submission.

This document follows the supplied single-column A4 article style, not IEEEtran's two-column layout. Final formatting must follow the faculty's confirmed requirements.